Matthew Middlehurst

Curriculum vitae

GitHub: MatthewMiddlehurst
LinkedIn: Matthew-Middlehurst
Google Scholar: Matthew Middlehurst

A dedicated researcher with a background in high-impact machine learning research. Experienced in open-source software engineering and mentoring student projects. Skilled in communication, with expertise in clearly conveying complex concepts. Interested in interdisciplinary collaboration and teaching the next generation of computer scientists using adaptable methods to support diverse needs.

Employment

Since 2025

Lecturer in Computer Science, *School of Computing and Engineering, University of Bradford*, Bradford, UK
A mixed research and teaching position. Module lead for the Fundamentals of Programming and Technical and Professional Skills modules to be taught at Yangzhou University (YZU).

2023–2025

Research Fellow, *School of Electronics & Computer Science (ECS), University of Southampton*, Southampton, UK

2022–2023

Senior Research Associate, *School of Computing Sciences (CMP), University of East Anglia*, Norwich, United Kingdom
A mixed research and software engineering role, developing the latest in machine learning algorithms for time series and maintaining the *aeon* open-source package. The position is part of EPSRC grant EP/W030756/2 where I contributed to the proposal. My role had the following responsibilities:

- Research time series machine learning techniques and applications
- Supervise project students at both Southampton and for the *aeon* project
- Develop framework and implementations based on time series literature
- Maintain the *aeon* project GitHub and community resources
- Facilitate meetings and communication for *aeon* developers and users, and connecting with multinational researchers

Education

2018–2022

PhD, Computer Science, *CMP, University of East Anglia*, Norwich, UK

Title Detection of Nuisance Call Centres using Improved Hybrid Time Series Classification Algorithms

Advisors Professor Gerard Parr & Professor Anthony Bagnall

Description My PhD thesis was a project developing state-of-the-art time series classification algorithms and applying them to detect scam/spam calls using data provided by British Telecom (BT).

2015–2018

BSc Computer Science (First-Class), *CMP, University of East Anglia*, Norwich, UK

Research Interests

Computer Science
- Time Series Machine Learning (Classification, Regression and Clustering)
- Real-time time series classification & Variable length series
- Healthcare & Human activity recognition applications
- Research Software Engineering

Publications and Conferences

Published highly cited works to high impact journals in DAMI, KAIS and Machine Learning, and have had my works accepted to competitive conferences such as IEEE Big Data and ECML-PKDD. Since my first publication in 2019, I have achieved a h-index score of 13 and my works have over 2281 collective citations according to Google Scholar.

Journal Articles

[J1] D. Guijo-Rubio, **M. Middlehurst**, G. Arcencio, D. F. Silva, and A. Bagnall, "Unsupervised feature based algorithms for time series extrinsic regression," *Data Mining and Knowledge Discovery*, vol. 38, no. 4, pp. 2141–2185, 2024.

[J2] C. Holder, **M. Middlehurst**, and A. Bagnall, "A review and evaluation of elastic distance functions for time series clustering," *Knowledge and Information Systems*, vol. 66, no. 2, pp. 765–809, 2024.

[J3] **M. Middlehurst**, A. Ismail-Fawaz, A. Guillaume, *et al.*, "Aeon: A python toolkit for learning from time series," *Journal of Machine Learning Research*, vol. 25, no. 289, pp. 1–10, 2024.

[J4] **M. Middlehurst**, P. Schäfer, and A. Bagnall, "Bake off redux: A review and experimental evaluation of recent time series classification algorithms," *Data Mining and Knowledge Discovery*, vol. 38, no. 4, pp. 1958–2031, 2024.

[J5] **M. Middlehurst**, J. Large, M. Flynn, *et al.*, "HIVE-COTE 2.0: A new meta ensemble for time series classification," *Machine Learning*, vol. 110, no. 11, pp. 3211–3243, 2021.

[J6] A. P. Ruiz, M. Flynn, J. Large, **M. Middlehurst**, and A. Bagnall, "The great multivariate time series classification bake off: A review and experimental evaluation of recent algorithmic advances," *Data Mining and Knowledge Discovery*, vol. 35, no. 2, pp. 401–449, 2021.

Conference Proceedings

[C1] **M. Middlehurst**, A. Bagnall, G. Forestier, A. Ismail-Fawaz, and A. Guillaume, "Time series machine learning with aeon: Classification and regression," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, Springer, 2025, pp. 432–437.

[C2] C. Qiu, **M. Middlehurst**, C. Holder, and A. Bagnall, "E-smote: A train set rebalancing algorithm for time series classification," in *International Workshop on Advanced Analytics and Learning on Temporal Data*, Springer, 2025, pp. 1–17.

[C3] A. Rushbrooke, **M. Middlehurst**, S. Sami, and A. Bagnall, "Channel selection and creation algorithms for electroencephalography classification with hive-cote," in *International Conference on Hybrid Artificial Intelligence Systems*, Springer, 2025, pp. 328–339.

[C4] A. Bagnall, **M. Middlehurst**, G. Forestier, *et al.*, "A hands-on introduction to time series classification and regression," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024, pp. 6410–6411.

[C5] **M. Middlehurst** and A. Bagnall, "Extracting features from random subseries: A hybrid pipeline for time series classification and extrinsic regression," in *International Workshop on Advanced Analytics and Learning on Temporal Data*, Springer, 2023, pp. 113–126.

[C6] **M. Middlehurst** and A. Bagnall, "The FreshPRINCE: A simple transformation based pipeline time series classifier," in *International Conference on Pattern Recognition and Artificial Intelligence*, Springer, 2022, pp. 150–161.

[C7] A. Bagnall, M. Flynn, J. Large, J. Lines, and **M. Middlehurst**, "On the usage and performance of the hierarchical vote collective of transformation-based ensembles version 1.0 (hive-cote v1. 0)," in *International Workshop on Advanced Analytics and Learning on Temporal Data*, Springer, 2020, pp. 3–18.

[C8] **M. Middlehurst**, J. Large, and A. Bagnall, "The canonical interval forest (CIF) classifier for time series classification," in *IEEE International Conference on Big Data (Big Data)*, IEEE, 2020, pp. 188–195.

[C9] **M. Middlehurst**, J. Large, G. Cawley, and A. Bagnall, "The temporal dictionary ensemble (TDE) classifier for time series classification," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, Springer, 2020, pp. 660–676.

[C10] **M. Middlehurst**, W. Vickers, and A. Bagnall, "Scalable dictionary classifiers for time series classification," in *International Conference on Intelligent Data Engineering and Automated Learning*, Springer, 2019, pp. 11–19.

Other

Conference Tutorials Co-led the applications for and presented two tutorials at the ECML-PKDD and KDD conferences in 2024. The ECML-PKDD tutorial provided an overview on the many time series learning tasks available in the *aeon* toolkit, while the KDD tutorial focused on classification and regression for time series data. Both tutorials were in collaboration with an international group of academics from multiple institutions. In 2025 held a tutorial at DSAA on time series classification for practical problems.

Teaching & Academic Experience

2025

Visiting Lecturer, *CMP, University of East Anglia*, Norwich, UK
Invited as a Visiting Lecturer to develop and present a lecture on time series data and machine learning for the Data Mining module in CMP. This will take place in April 2025.

Since 2024

Student Project Supervisor, *ECS, University of Southampton*, Southampton, UK
Secondary supervisor to two PhD students who started in 2024, and secondary supervisor to multiple MSc students for their MSc projects.

Since 2024

Google Summer of Code Mentor
Took part in the Google Summer of Code 2024 and 2025 as a mentor and administrator for the *aeon* project. I mentored three undergraduate level student projects for the summer, guiding them in how to effectively contribute to open source packages and implement time series algorithms. As part of this process I designed projects for prospective students to apply for and processed applicants.

2018–2022 **Associate Tutor**, *CMP, University of East Anglia*, Norwich, UK
Taught on multiple modules to undergraduate and postgraduate students in both in-person and remote environments. I taught on the following modules: Programming, Data Structures and Algorithms and Machine Learning. I contributed to the design of assessment and marking for these modules.

2019–2020 **HE Teaching Skills Course**, *CMP, University of East Anglia*, Norwich, UK
Completed a course on developing teaching skills while working as an Associate Tutor. The module focused on the development of practical skills and confidence in classroom practice in higher education teaching. The module was accredited by the Higher Education Academy (HEA) and aligned with the UK Professional Standards Framework (UKPSF) for teaching and supporting learning in higher education at Descriptor 1 level.

Since 2023 **Academic peer-review**
Since 2023 I have reviewed submissions for the ECML-PKDD conference and the attached AALTD workshop. I have peer-reviewed for other venues such as KDD on occasion.

Skills

Programming & Technical Java, Python, Git, CI/CD, Software Engineering

Academic Machine Learning, Mentoring, Research Publication, Academic Writing

Professional Teamwork, Communication, Public Speaking

Other Relevant Professional Experience

Since 2021 **Grant Writing**
My research and development of *aeon* led to me writing the proposal for EPSRC grant EP/W030756/2 with my supervisor, which funds my current postdoctoral position. The grant was one of 11 funded out of 254 proposals as part of a special call. Building on my HIVECOTE 2.0 publication, I helped write a grant on explainable and sustainable time series machine learning with my supervisor Anthony Bagnall. It is submitted to EPSRC responsive mode, and if funded I will lead a work package.

2024–2025 **LinusBio Consultancy**, *https://www.linusbio.com/*
Consulted with American tech company LinusBIO relating to their ClearStrandASD product. We are working to find whether time series machine learning algorithms can detect biomarkers associated with autism. This is organised through the University of Southampton, and LinusBIO are providing funding in the amount of £16,000.

Since 2018 ***aeon* Core Developer**, *https://github.com/aeon-toolkit/aeon*
Core Developer for the Python based *aeon* open-source toolkit (formerly *sktime*) for time series analysis. My activities in the project involve managing pull requests, creating code and documentation, maintaining CI/CD and developing new framework. I am involved in the management of the project, and helped lead the application which had it accepted as a NumFOCUS affiliated project.